FDS EXPERIMENT-9

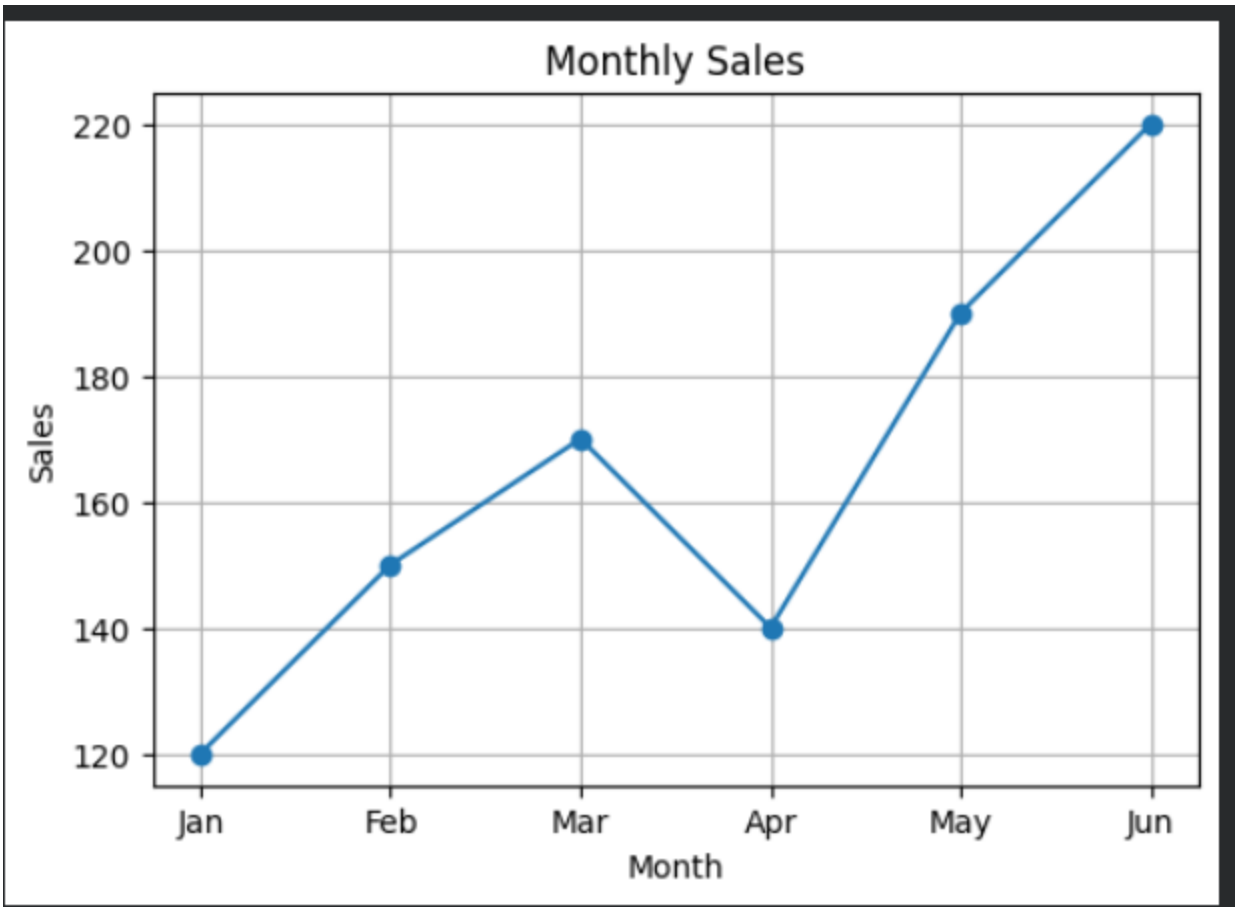
SUBIN P – 250701750

Experiment to understand Matplotlib library use cases in Data Science through visualization.

```
import matplotlib.pyplot as plt
import numpy as np
```

```
# Sample data
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [120, 150, 170, 140, 190, 220]
```

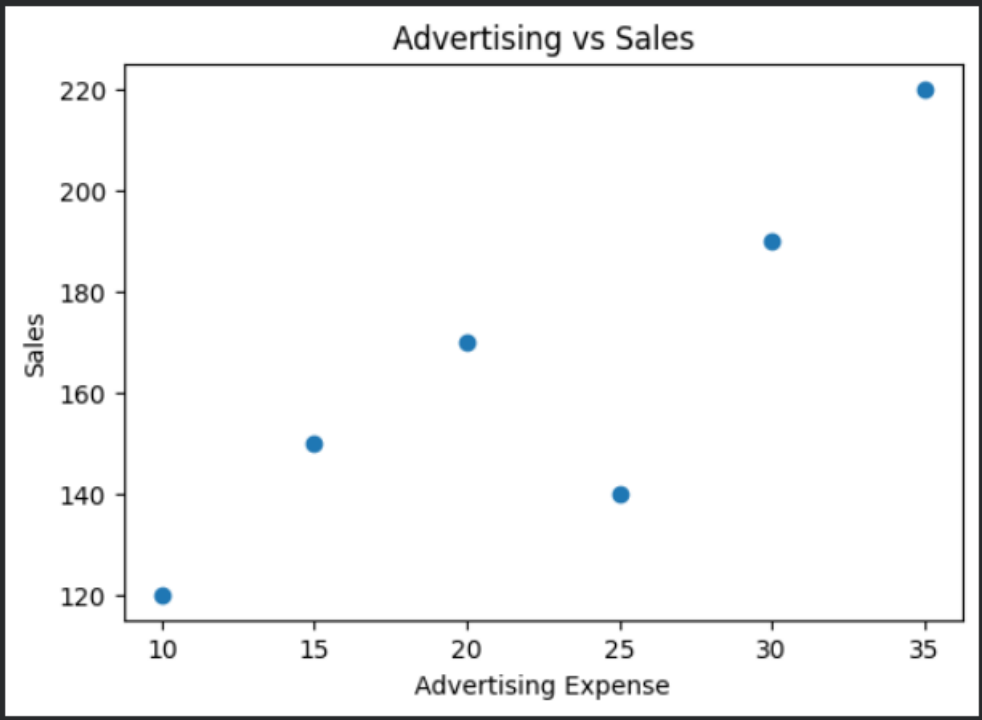
```
# 1. Line Plot
plt.figure(figsize=(6, 4))
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```


Monthly Sales
220
200
180
160
140
120
Sales
Jan
Feb
Mar
Apr
May
Jun
Month

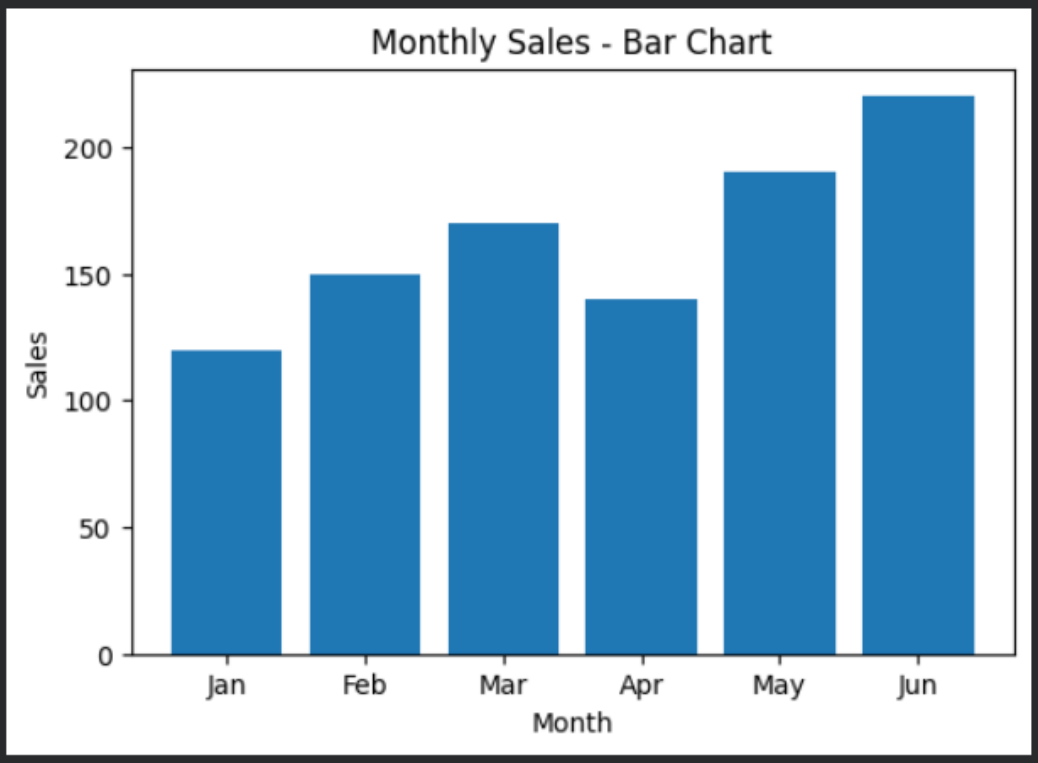
```
[ ]
```

```python
# 3. Scatter Plot
advertising = [10, 15, 20, 25, 30, 35]

plt.figure(figsize=(6, 4))
plt.scatter(advertising, sales)
plt.title("Advertising vs Sales")
plt.xlabel("Advertising Expense")
plt.ylabel("Sales")
plt.show()
```

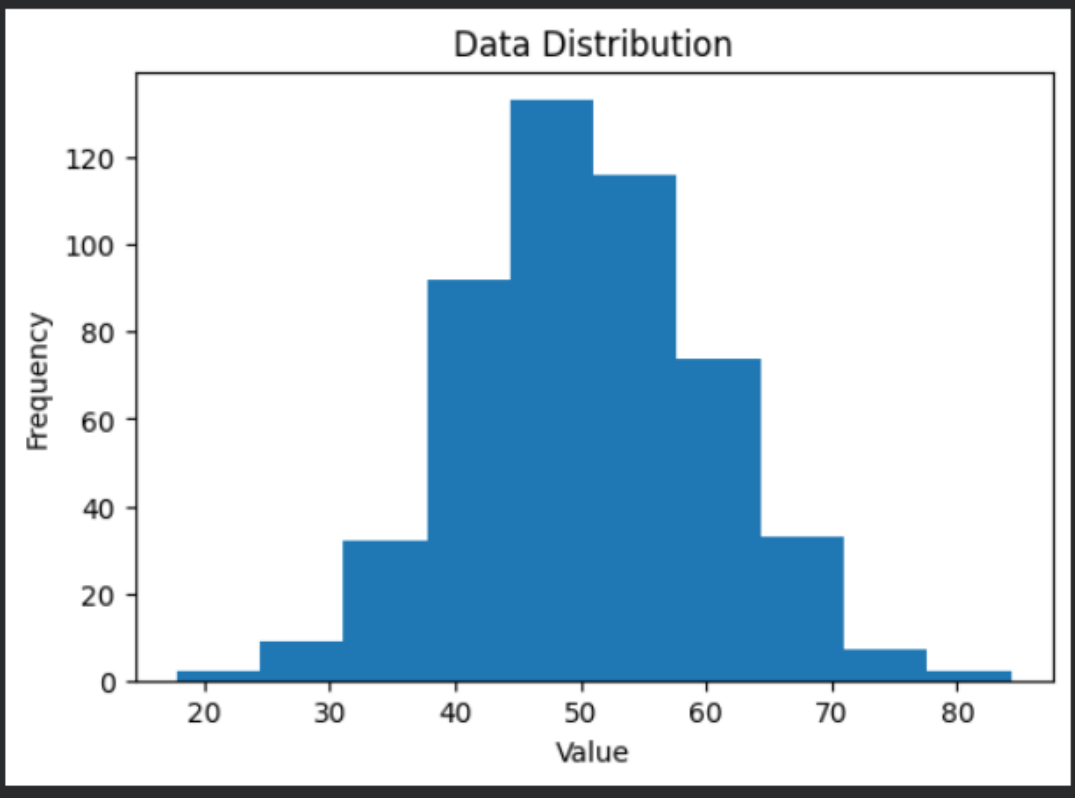


```python
# 2. Bar Chart
plt.figure(figsize=(6, 4))
plt.bar(months, sales)
plt.title("Monthly Sales - Bar Chart")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```



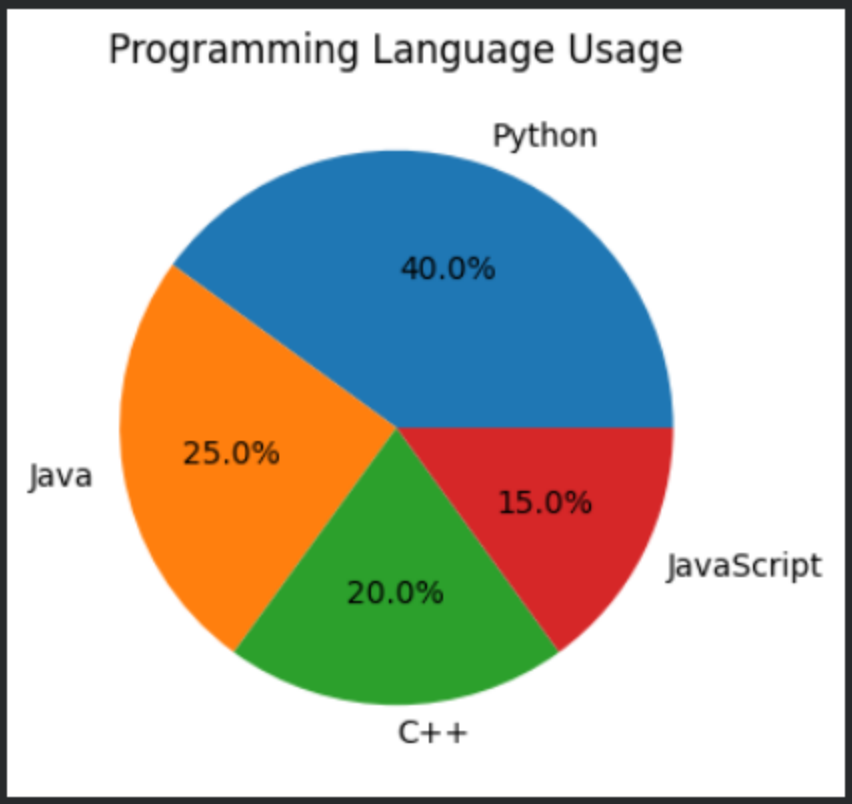
```python
# 4. Histogram
data = np.random.normal(50, 10, 500)

plt.figure(figsize=(6, 4))
plt.hist(data, bins=10)
plt.title("Data Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```



```python
# 5. Pie Chart
labels = ['Python', 'Java', 'C++', 'JavaScript']
values = [40, 25, 20, 15]

plt.figure(figsize=(6, 4))
plt.pie(values, labels=labels, autopct='%1.1f%%')
plt.title("Programming Language Usage")
plt.show()
```



```python
# 6. Box Plot
data = [12, 15, 14, 10, 18, 20, 16, 15, 14, 50]

plt.figure(figsize=(6, 4))
plt.boxplot(data)
plt.title("Box Plot")
plt.ylabel("Values")
plt.show()
```

